

MOBILOITTE TECHNOLOGIES

Enterprise AI, RAG & Web3 Technical Implementation Blueprint

Governed AI. Built for Business. A Deep Technical Guide to Architecting Production-Grade Retrieval-Augmented Generation, Agentic Workflows, and Zero-Trust Enterprise Integrations.

Document Type: Technical Capability & Architecture Reference

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1. Executive AI Strategy & System Architecture

Mobiloitte Technologies leads enterprise digital engineering by bridging foundational artificial intelligence models with production-ready business logic. This Blueprint provides a reference framework for deploying Governed AI, Retrieval-Augmented Generation (RAG) engines, Agentic AI, and secure Blockchain infrastructures at enterprise scale.

1.1 Core Architectural Principles

Mobiloitte Governed AI Engineering Principles:

- **Source Grounding:** All generative responses must be deterministically linked to validated internal enterprise knowledge bases via high-density vector indices to eliminate hallucinations.
- **Privacy by Design:** Enterprise data ingested for RAG indexing or LLM fine-tuning is isolated in zero-egress VPC environments, ensuring zero leakage to public foundational models.
- **Multi-Agent Orchestration:** Complex enterprise tasks are decomposed into specialized tool-calling agents equipped with strict reasoning cores and memory systems.
- **Immutable Audit Trails:** Transactional events, model outputs, and automated agent actions are logged to cryptographic ledgers for compliance and governance auditing.

1.2 Enterprise AI Technology Matrix

Architecture Layer	Core Stack / Tools	Enterprise Purpose & Capability
Foundational Models & APIs	GPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, Llama 3 (Fine-Tuned)	High-reasoning core engines, multimodal document parsing, and domain-specific logic.
RAG & Vector Storage	Pinecone, ChromaDB, Qdrant, LlamaIndex, LangChain	Dense semantic text embeddings, 500-1000 char chunking, real-time vector similarity search.
Agentic Orchestration	CrewAI, LangGraph, AutoGen, n8n, Zapier AI	Multi-agent collaboration, stateful reasoning loops, tool-calling REST integrations.
Bot & Multi-Channel Deployment	ConveriqoAI BOT Platform (Proprietary)	Unified omni-channel deployment across Web, WhatsApp, iOS/Android, IVR, and ERP/CRM.
DevOps & MLOps Pipelines	Docker, Kubernetes (EKS), GitHub Actions, Terraform, MLflow	Continuous integration, automated model re-training, containerized low-latency serving.

2. RAG Pipeline Engineering & Agentic Workflows

2.1 Technical RAG Data Ingestion Pipeline

Semantic Chunking & Embedding Standards

Mobiloitte's proprietary RAG ingestion engine converts raw, unstructured enterprise documents (PDFs, Word documents, Confluence pages, ERP records) into high-dimensional vector embeddings designed for rapid, low-latency semantic retrieval.

- **Text Splitting Parameters:** Documents are systematically chunked into target sizes of **500 to 1,000 characters**.
- **Contextual Overlap Ratio:** Enforcing a mandatory **10% to 20% overlap** between adjacent chunks to preserve semantic context across chunk boundaries.
- **Embedding Model:** Utilizing OpenAI text-embedding-3-large or open-source BGE/E5 embeddings normalized for cosine similarity distance.
- **Hybrid Search Engine:** Combining dense vector search with sparse keyword search (BM25) and re-ranking algorithms (Cohere Rerank) to maximize retrieval precision.

2.2 Agentic Multi-Agent Framework

Tool-Calling Agent Architecture Pattern

Rather than relying on single monolithic LLM prompts, Mobiloitte deploys specialized multi-agent networks powered by CrewAI and LangGraph.

Research Agent Validation Agent Tool Executor Formatting Agent Security Auditor

- **Supervisor Node:** Coordinates task delegation, evaluates agent outputs against business rules, and triggers fallback routines.
- **Tool-Calling Integration Layer:** Equips agents with authenticated REST/GraphQL API access to internal CRM (Salesforce, HubSpot), ERP (SAP), and SQL databases.
- **Human-in-the-Loop (HITL):** Mandates human sign-off for high-risk transactional steps (financial approvals, medical diagnostics, contract execution).

2.3 The ConveriqoAI BOT Platform

Mobiloitte's flagship enterprise product—the **ConveriqoAI BOT Platform**—provides a unified interface for configuring, governing, and deploying agentic bots across communication channels.

- **Omni-Channel Connectors:** Native support for Web Chat, WhatsApp Business API, Mobile SDKs (iOS/Android), and Voice IVR integrations.
- **Built-in RAG Assistant:** Direct connector hooks into internal document stores with real-time source attribution for end users.
- **Enterprise Role-Based Access Control (RBAC):** Granular permissions ensuring users only retrieve information authorized for their corporate access level.

3. Web3, Cloud Infrastructure & Cybersecurity

3.1 Enterprise Blockchain & Tokenization Architecture

Ledger Infrastructure & Smart Contract Security

Mobiloitte engineers private, public, and hybrid blockchain implementations to drive supply chain transparency, asset tokenization, and decentralized finance protocols.

- **Permissioned Enterprise Chains:** Hyperledger Fabric setups for consortiums in healthcare, trade finance, and logistics.
- **Public Chain Smart Contracts:** Production EVM deployment in Solidity and Rust on Ethereum, Polygon, and Solana.
- **Real-World Asset (RWA) Tokenization:** Industrial-grade tokenization of real estate, commodities, and financial instruments with built-in KYC/AML identity hooks.
- **Smart Contract VAPT Auditing:** Automated static analysis (Slither, Mythril) combined with manual formal verification to guarantee zero exploit vulnerabilities prior to mainnet deployment.

3.2 Zero-Trust Cybersecurity & Compliance Matrix

Security is natively embedded into every layer of Mobiloitte's software development life cycle (SDLC).

Security Vector	Implementation Standard	Regulatory Alignment
Data Encryption	TLS 1.3 in transit, AES-256 at rest across all database clusters.	GDPR, ISO 27001
Access Control	OAuth 2.0, OpenID Connect, Multi-Factor Authentication (MFA), Granular RBAC.	SOC 2 Type II, HIPAA
Vulnerability Testing	Continuous OWASP Top 10 automated scanning & bi-annual third-party penetration testing.	PCI-DSS, NIST
Data Sovereignty	Local cloud region hosting (e.g., UAE AWS region for Middle East PDPL compliance).	UAE PDPL, POPIA (South Africa)

3.3 Featured Enterprise Case Studies & Metrics

- **IFFCO Kisan (Agritech Marketplace):** Connected 1,000,000+ farmers with real-time crop intelligence, driving a 30% crop yield improvement.
- **Nesba (PropTech Platform):** Scaled a fractional real estate platform in the MENA region to 50,000+ active investors with a 15% average ROI.
- **Steve Madden (Lifestyle App):** Modernized mobility applications achieving over 2,000,000+ downloads and a 65% user retention rate.
- **Bitage (Fintech Trading Automation):** Processed \$10,000,000+ in transactional volume with 94% algorithmic execution accuracy.

4. Engagement Models & Global Contact Directory

4.1 Flexible Client Engagement Models

Mobiloitte Partnership Frameworks:

- **Dedicated Engineering Squads:** Full-stack, cross-functional engineering pods (Architects, Developers, QA, DevOps) integrated directly into client agile teams.
- **15-Day MVP Accelerator Sprint:** Rapid prototyping framework leveraging pre-built accelerators, Vite scaffolding, and AI coding tools for quick product validation.
- **Fixed-Price Project Delivery:** Milestone-based delivery for structured projects with clearly defined software specifications and SLA guarantees.

4.2 Global Office Directory & Contact Points

Regional Office	Physical Location Address	Contact Information
India (Global HQ)	D-115, Okhla Phase-1, New Delhi - 110020, India	Phone: +91-9999525801 Email: connect@mobiloitte.com
United States	Mobiloitte Inc., 1302 Lords Ct, Wilmington, MA 01887, USA	Phone: +1-7816706589 Email: sales@mobiloitte.com
United Kingdom	Suite 415c, Margaret Powell House, Midsummer Blvd, Milton Keynes MK9 3BN	Phone: 1800-5691801 Email: sales@mobiloitte.com
United Arab Emirates	AL DANA CENTRE, Plot No. 702-0, Al Muraqabat, Dubai, UAE	Phone: +971-56329600 Email: connect@mobiloitte.com
Singapore	1 Raffles Place, #34-04, One Raffles Place, Singapore 048616	Phone: 1800-5691801 Email: sales@mobiloitte.com
South Africa	8 Ashwood Drive, Centurion 0185, South Africa	Phone: 1800-5691801 Email: hr@mobiloitte.com

4.3 Key Contact Emails

- **General Inquiries:** connect@mobiloitte.com
- **Enterprise Sales & RFPs:** sales@mobiloitte.com
- **HR Operations & Recruitment:** hr@mobiloitte.com (Extension Line: 204)
- **Official Website:** www.mobiloitte.com